Lesson notes · for you

The sensors in your phone

IoT · Level 1 · Lesson 2

Materials
A phone with phyphox, a compass app or a sound-meter app (with a parent's OK); Paper and a pencil for your table; No phone: squared paper and a pencil

Learning goals

- Name at least four sensors in a phone and what each is for.
- Explain that a camera is millions of light sensors that make a grid of numbers.
- Measure something with a phone sensor app.

Activity: Measure with your phone

1. Ask a parent or guardian first. Install phyphox (free) from the official app store, or use the compass app already on the phone.
2. Pick light, loudness or the compass direction. Light works in phyphox on most Android phones, not on iPhones.
3. Choose two places, like by a window and at the back of the room, or the classroom and the library. Hold the phone the same way each time.
4. Make a table: place, readings 1, 2 and 3, and the unit (lux for light, decibels for loudness, degrees for direction).
5. Which place had the bigger number, and why? Write two sentences. No phone: make a 10 × 10 number picture on squared paper instead.

Ask before installing anything, and use only the official app store. Light readings work on most Android phones, not on iPhones. No phone? The number picture on squared paper teaches the camera's idea just as well.

Discussion

- Which sensor would you miss most if your phone lost it?
- Why might two phones give different numbers in the same place?
- What could the sensors in your phone tell an app about your day?

Check yourself

- Name four phone sensors and one job for each.

Try it this week
With a parent's OK, measure one thing in two places with your phone, or make a number picture on squared paper.

Worksheet

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Name _____ Class _____ Date _____

1 **You take a photo of the school garden with a 12-megapixel camera. What does the phone actually store?**

- ◯ About 12 million pixels, each stored as three numbers
- ◯ A tiny copy of the scene, like a printed photo inside the phone
- ◯ One number for the whole photo: how bright the garden was

2 **Match what your phone does to the sensor behind it.**

The screen turns when you turn the phone	• • • •	Light sensor
The screen goes dark at your ear in a call	• • • •	Accelerometer (motion sensor)
The screen dims in a dark room	• • • •	Compass sensor
Your map faces the way you're walking	• • • •	GPS receiver
Your position shows on a saved map, with no data	• • • •	Proximity sensor

3 **Your phone's GPS needs mobile data or Wi-Fi to work out where you are.**

◯ True ◯ False

4 **Work out how many numbers a photo holds.**

A 12-megapixel photo has about _____ million pixels. With 3 numbers per pixel (red, green, blue), that's about _____ million numbers.

Word bank: 36 · 15 · 12 · 4

5 **Amani's screen stays on during calls, and her cheek keeps pressing buttons. What could be the cause?**

- ☐ The light sensor is broken
- ☐ A thick screen protector covers the proximity sensor
- ☐ The microphone is too quiet
- ☐ The GPS is switched off
- ☐ The proximity sensor is dirty or broken

✎ **Which phone sensor surprised you most? What could it tell an app about you if the app used it without asking?**

Check your answers

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1 **About 12 million pixels, each stored as three numbers**

Each of the camera's millions of light sensors gives numbers, and every pixel is stored as about three numbers: red, green and blue.

2 **The screen turns when you turn the phone — Accelerometer (motion sensor)**
The screen goes dark at your ear in a call — Proximity sensor
The screen dims in a dark room — Light sensor
Your map faces the way you're walking — Compass sensor
Your position shows on a saved map, with no data — GPS receiver

Each job belongs to one sensor. The motion sensor knows which way is down, the proximity sensor notices your ear, the light sensor measures brightness, the compass feels magnetism, and the GPS listens to satellites.

3 **False**

The GPS receiver listens to satellites, so it finds your position with no data at all.

4 **A 12-megapixel photo has about 12 million pixels. With 3 numbers per pixel (red, green, blue), that's about 36 million numbers.**

Mega means million, so 12 megapixels is about 12 million pixels, and 12 × 3 = 36 million numbers.

5 **A thick screen protector covers the proximity sensor; The proximity sensor is dirty or broken**

Switching the screen off at your ear is the proximity sensor's job, so anything that blocks or breaks it causes this.